Delhi 2024 LULC Classification and Validation Report

Purpose

This report documents preparation and validation of the 2024 Delhi land-use/land-cover (LULC) dataset prepared for subsequent Markov-chain analysis. The final product is a categorical 10 m GeoTIFF.

1. Final Raster Properties

Property	Value
File	Delhi_LULC_2024_FINAL_NODATA.tif
Format	GeoTIFF (GTiff)
Resolution	10 m × 10 m
CRS	EPSG:32643 – WGS 84 / UTM zone 43N
Dimensions	5028 × 5363 pixels
Bands	1 categorical band
Data type	Byte / UInt8
NoData	255 (outside study area)

2. Class Codebook

Code	Class
0	Water
1	Trees
2	Grass
3	Flooded vegetation
4	Crops
5	Shrub & scrub
6	Built-up
7	Bare ground
8	Snow & ice
255	NoData / outside study area

3. Validation Method

A total of 240 validation points were visually checked against reference imagery. The reference class was recorded independently and compared with the Dynamic World class at each point. The final validation dataset contains two disagreements, which were retained rather than forcing agreement.

4. Confusion Matrix (Rows = Reference; Columns = Dynamic World)

Ref.\DW	0	1	2	3	4	5	6	7
0 Water	29	0	0	0	0	0	0	0
1 Trees	0	30	0	0	0	0	0	0
2 Grass	0	0	30	0	0	0	0	0
3 Fl. veg.	0	0	0	30	0	0	0	0
4 Crops	1	0	0	0	30	0	0	0

Ref.\DW	0	1	2	3	4	5	6	7
5 Shrub	0	0	0	0	0	30	0	0
6 Built-up	0	0	0	0	0	0	30	1
7 Bare	0	0	0	0	0	0	0	29

Class 8 (Snow & ice) occurs in only 22 raster pixels and was not represented among the 240 validation points; therefore it is not part of the accuracy matrix.

5. Accuracy Results

Metric	Result
Validation samples	240
Correct classifications	238
Misclassifications	2
Overall Accuracy	99.17%
Cohen's Kappa	0.9905

6. Observed Errors

Two off-diagonal observations were identified: one reference Crops point was classified as Water, and one reference Built-up point was classified as Bare ground. These isolated disagreements are plausible at mixed or spectrally similar pixels, particularly around crop/water and built-up/bare-ground boundaries.

7. Area Statistics

Code	Class	Pixels	Area (ha)	Area (km²)
0	Water	268,160	2,681.60	26.8160
1	Trees	1,599,430	15,994.30	159.9430
2	Grass	4,903	49.03	0.4903
3	Flooded vegetation	10,228	102.28	1.0228
4	Crops	3,947,171	39,471.71	394.7171
5	Shrub & scrub	88,248	882.48	8.8248
6	Built-up	8,587,457	85,874.57	858.7457
7	Bare ground	532,740	5,327.40	53.2740
8	Snow & ice	22	0.22	0.0022

8. Validation Statement for Reporting

The 2024 LULC classification achieved 99.17% overall accuracy on the 240-point validation sample, with a Cohen's kappa coefficient of 0.9905. Two misclassifications were observed: Crops to Water and Built-up to Bare ground. The results indicate very strong agreement between the classification and the independently interpreted validation sample. These accuracy values represent the sampled validation assessment and should not be interpreted as exhaustive pixel-wise accuracy for the entire Delhi study area.

9. Handover Package

The Markov-analysis package should contain the final GeoTIFF, final validation CSV, area-statistics Excel file, and class-codebook text file. The GeoTIFF is the primary Markov input; the other files provide documentation and quality-control information.